Grasping Section of the Team Description Paper (Draft)

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RoboCup@Home 2027

1 Dexterous Grasping

1.1 Learning Grasps for a Low-Cost Servo Hand

pib's 3D-printed hand has eight bus servos and force sensors in the fingertips, but no camera or torque sensing. We train a grasp policy that uses only joint angles and fingertip forces with PPO in Isaac Lab [1] on a digital twin of the hand, randomizing object shape, mass and friction. Cheap servos wear out when they jitter and shut down when they push against their torque limit for too long. We therefore smooth the policy with CAPS [2] and limit the time servos spend near ($\geq 80\,\%$) their stall torque with a PID Lagrangian [3]. In simulation (five seeds), the policy holds 83 % of the benchmark objects (scripted grasp: 69 %) and 74–82 % of unseen YCB objects [4], while servos spend 13 % instead of 32 % of the time near stall. Code and all trained policies are open source[1]; tests on the real hand are pending.

References

1. Mayank Mittal, Pascal Roth, James Tigue, Antoine Richard, Octi Zhang, et al. Isaac Lab: A GPU-accelerated simulation framework for multi-modal robot learning. *arXiv preprint arXiv:2511.04831*, 2025.
2. Siddharth Mysore, Bassel Mabsout, Renato Mancuso, and Kate Saenko. Regularizing action policies for smooth control with reinforcement learning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 1810–1816, 2021.
3. Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by PID Lagrangian methods. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, volume 119 of *PMLR*, pages 9133–9143, 2020.
4. Berk Calli, Arjun Singh, Aaron Walsman, Siddhartha Srinivasa, Pieter Abbeel, and Aaron M. Dollar. The YCB object and model set: Towards common benchmarks for manipulation research. In *International Conference on Advanced Robotics (ICAR)*, pages 510–517, 2015.

[1] `https://github.com/churro206/pib-grasp-sim`